

AI Job Finder: Hardened SOTA Implementation Plan

Target Root: D:\Ai job finder | Architecture: 2026 SOTA Local-First Copilot | Status: Ready for Execution

1. Executive Tooling Scorecard & Upgrade Matrix

Subsystem	Previous Choice	SOTA Replacement	Score	Core Architectural Rationale
Vector DB	ChromaDB (embedded)	LanceDB	5.5 → 9.5	Eliminates SQLite write-lock timeouts during concurrent ingestion; zero-copy mmap.
Embedding	all-MiniLM-L6-v2	BAAI/bge-small-en-v1.5	5.5 → 9.0	512-token context prevents truncating 70% of job requirements; +11.25 MTEB gain.
Lexical Search	rank-bm25	bm25s (disk mmap)	5.0 → 9.5	Sparse matrix precomputed index; 500x faster execution; zero boot re-tokenization.
PDF Engine	Playwright Chromium	Typst / RenderCV v2	7.5 → 9.5	Compiles in 15ms; avoids Chromium Skia kerning fragmentation; guaranteed ATS text order.
Stealth Browser	Camoufox + Playwright	Camoufox Hardened	8.5 → 9.5	C++ Gecko spoofing immune to JS prototype inspection; enable humanize & geoip.
Task Queue	SqliteHuey	SqliteHuey (WAL Mode)	9.0 → 9.5	Windows native zero-daemon queue; enable WAL and busy_timeout=5000.
Frontend UI	React 19 + Vite 8	React 19 + Virtualizer	9.0 → 9.5	Add @tanstack/react-virtual to limit active DOM cards to ~15 for locked 60 FPS.
API Backend	FastAPI + uvicorn	FastAPI Lifespan	9.5 → 9.8	Eagerly preload models in lifespan handler; enable GZipMiddleware for fast transfers.

Phase 0: Security Hardening & Execution Bug Fixes

Task 0.1: Fix Async Desync in Browser Agent (D:\Ai job finder\src\browser_agent.py:210)

Resolve silent autofill failure where async Playwright page was passed to sync runner. Convert to `_fill_page_async` with full locator actionability checks.

Task 0.2: Eliminate Firefox Profile Collision Traps (D:\Ai job finder\src\stealth_browser.py)

Enforce isolated profile directory allocation (`data/profiles/worker_{id}`) to prevent Firefox parent.lock crashes under concurrent scraping.

Task 0.3: Scrub Hardcoded Secrets (D:\Ai job finder\src\config.py)

Ensure all passwords, API tokens, and session secrets load strictly via python-dotenv from `.env`.

Verification Gate 0: `pytest tests/test_browser_agent.py` → PASS with 0 unawaited coroutine warnings.

Phase 1: Vector DB & Embedding SOTA Migration

Task 1.1: Install Dependencies: `pip install lancedb sentence-transformers==3.3.0 pyarrow`

Task 1.2: Upgrade Matcher Model (D:\Ai job finder\src\matcher.py)

Replace all-MiniLM-L6-v2 with BAAI/bge-small-en-v1.5. Preserves identical 384-dim structure while doubling context to 512 tokens to capture full job qualifications.

Task 1.3: Build Migration Script (D:\Ai job finder\scripts\migrate_to_lancedb.py)

Read records from `jobfinder.db`, compute embeddings, and initialize LanceDB table at `data/lancedb`.

Verification Gate 1: Execute migration script; verify LanceDB directory populated and vector search returns without database lock errors.

Phase 2: Lexical Search Upgrade (rank-bm25 → bm25s)

Task 2.1: Install bm25s: pip install bm25s PyStemmer

Task 2.2: Implement Persistent bm25s Indexer (D:\Ai job finder\src\matcher.py)

Precompute sparse indices and save to disk with memory mapping (mmap=True), cutting query latency to <2ms without in-memory retokenization on server boot.

Task 2.3: Maintain Reciprocal Rank Fusion (RRF): Retain scale-invariant RRF (k=60) combining LanceDB dense rankings and bm25s lexical rankings.

```
Verification Gate 2: python -c "from src.matcher import query_bm25; docs, s = query_bm25('python', 5);
assert len(docs) > 0"
```

Phase 3: High-Fidelity ATS PDF Engine (Typst / RenderCV)

Task 3.1: Integrate Typst Document Compilation (D:\Ai job finder\src\pdf_engine.py)

Generate clean .typ markup and invoke typst compile. Guarantees 100% linear text extraction and standard CMaps for taleo/workday ATS systems.

Task 3.2: Relegate Playwright PDF to Visual Preview: Use Chromium HTML-to-PDF strictly for in-browser visual previews; all job submissions compile via Typst.

```
Verification Gate 3: Extract text via pypdf from output PDF; verify unbroken word boundaries and zero
kerning artifacts.
```

Phase 4: Browser Stealth Hardening & Huey Concurrency

Task 4.1: Inject Camoufox Anti-Detect Parameters (D:\Ai job finder\src\stealth_browser.py)

Enable humanize=True (Bézier mouse curves and typing jitter) and geoip=True (automatic timezone and WebRTC alignment).

Task 4.2: Enforce WAL Mode on Huey Queue (D:\Ai job finder\src\tasks.py)

Set journal_mode=wal and busy_timeout=5000 in SqliteHuey pragmas.

Task 4.3: Threaded Consumer Execution Script: Update run_worker.bat with -k thread -w 2.

```
Verification Gate 4: Start worker; verify data/huey.db-wal exists and background tasks execute without
locking web server.
```

Phase 5: Frontend DOM Virtualization & Server Lifespan Preloading

Task 5.1: Virtualize Job Feed (D:\Ai job finder\frontend): Install @tanstack/react-virtual and wrap LiveIntelFeed.tsx list, capping rendered nodes to ~15 active cards.

Task 5.2: Configure Server Lifespan Preloading & GZip (D:\Ai job finder\server.py): Preload SentenceTransformer off the HTTP request path; add GZipMiddleware.

```
Verification Gate 5: npm run build clean; API response latency under 50ms with GZip compression.
```